Strategic Insights from Technical to Management Level: Private Hospital Finance

A Work by:

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Contents

Introduction

Data visualization is a very important tool in modern data analysis because it helps people understand complex information quickly and in a more intuitive way. Instead of reading large tables or raw numbers, visualizations like charts and dashboards allow users to identify trends, patterns, and outliers much faster. This is especially important in business environments where decisions must be taken quickly and based on clear insights. Without proper visualization, even correct data can be confusing and difficult to explain to others, which reduces its actual value.

Of course, the above paragraph is ideal. It describes the importance of data visualizations without taking into account the necessity of preprocessing using the unlimited capabilities of Python. According to the authors, in real-life scenarios data preprocessing and exploratory data analysis (EDA) may overlap, as data preprocessing by itself reveals much information about the dataset, even prior to applying EDA. In a more professional way, it can be assumed that data visualizations are dashboards made for quick strategic KPI communication with the stakeholders, and they always require data preprocessing and some exploratory data analysis prior to their creation.

Patient discharges is a critical topic in healthcare systems. It is directly related with hospital efficiency, operational costs and also the quality of care that patients receive. For example, long length of stay may indicate inefficiencies, while very short stays may hide other issues. In the dataset, costs, revenues and patient flows are explored across different hospitals and years, trying to identify patterns and possible problems. Also, differences between hospitals can show how each one manages its resources, although the data does not always explain the reasons behind these differences

In this project, the results are presented across different levels of management hierarchy. The technical level focuses on detailed exploration and metrics, while the director level highlights comparisons and performance. Finally, the C-level view is more simplified and focused only on the most important KPIs and general picture.

1. Data Source Selection

We obtained the data from the **SPARCS NYS** (Statewide Planning and Research Cooperative System) for the years 2018-2019.

It consists of 100,000+ patient encounters, de-identified with Clinical and Financial data of the Business as Usual model of operation. The volume provides statistical power to ensure the analysis is robust and the maxima are not really outliers but rather the hospital system realities relevant to the operation model of the facility.

The years were selected to be before the Covid-19 period to ensure we do not have anomalies that we will not be able to understand and analyse.

2. Data Preprocessing & Exploratory Data Analysis

To prepare the dataset for the dashboard analysis we performed Data Cleaning and Integrity procedures:

2.1. Feature selection on Business Grounds

The initial dataset consisted of 33 features. Some of them being too detailed on the illness specifics of the sample. Since these required deep domain knowledge, we excluded most of them and decided to base our analysis on the features listed in the following table.

Table 1: The features selected from the initial dataset and on which we based our analysis

Clinical Features	**Facility Name**	The name of the hospital providing health services . We have downloaded data for 3 NY state hospitals. **Maimonides, Hudson Valley, St. Joseph's**
	Age Group	Age in years at time of discharge. Grouped into bins (0 to 17, 18 to 29, 30 to 49, 50 to 69, and 70 or Older)
	Length of Stay (LOS)	The total number of patient days per row of the dataset
	Severity of Illness	Minor (1), Moderate (2), Major (3), Extreme (4)
	Admission Type	Elective, Emergency, Newborn, Trauma, Urgent

	Diagnosis	Illness categorization for invoicing
	ED Indicator	If Emergency Department (ED) was invoiced
	Payment Type	**Medicare** Federal covered Insurance for ages > 65 and for younger people with certain disabilities **Medicaid** Joint Federal and State program for people with limited income and resources **Private Health Insurance** Coverage provided by commercial entities (like Blue Cross, Aetna, or UnitedHealthcare) often through an employer **Self-Pay / Other** Includes patients who pay out-of-pocket
Financial Features	**(Total) Costs**	Estimated Real Cost of a discharge. Internal to the hospital
	(Total) Charges	Estimated Total amount to be charged to the patient according to internal and Insurance policies.

2.2. Feature Engineering

Table 2:The features engineered to facilitate the analysis

Engineered Features	**Charge-to-Cost Ratio (CCR)**	Charges / Costs
	Markup (Profit)	Charges - Costs
	Markup % **Profit-to-Cost**	(Charges - Costs)/Costs×100
	Costs per Patient per Day (CPD)	Costs / Patients x LOS
	Revenue per Patient per Day (RPD)	Charges / Patients x LOS
	CPD to RPD	COST / Charges (per Patient per Day)
	Cross Subsidy Ratio (CSR)	Profit _CashCows / Profit_LossMakers
	Categorical Re-mapping	Severity of Illness - Ranking Age - Ranking

2.3. Integrity Checks

We performed a check of data types, relating to the business content of the features

- Corrected the Costs and Charges to be numeric values, where the $ sign mark was preventing it from being a number, which was necessary as these features are used for numerical calculations.
- Resolved the 120+ LOS issue. Again this feature could not be defined as a number. For length of stay > 120 days values are defined as “120+”. So, we assigned those fields the number 120.

2.4. Handling Nulls & Duplicates

The dataset exhibited exceptional cleanliness, with a missing value/NaN rate of only **0.003%**, allowing us to retain nearly the entire population for analysis.

Duplicates existed in the dataset from the beginning. We explained it as cases that had the same exact symptoms and invoicing details. Their number increased after feature selection, since the granularity that was created by some of the excluded fields was eliminated after their deletion. Additionally, there was no field that would account for a patient id. So, we did not consider anything as a duplicate. We considered everything to be a correct de-identified input.

2.5. Feature Exclusion

Columns with low variance were identified, but we decided there was no meaning in excluding them, since even with 2 value counts, they would give the information of male, female for example or having an ED Indicator or not, which is valuable.

By establishing this baseline, we ensure that the subsequent findings coming from our analysis are not products of data instability, but accurate reflections of the hospitals’ pre-pandemic organizational structure and operational practices.

2.6. Univariate Analysis

2.6.1. Numeric Features

2.6.1.1. Summary Statistics

```
# Summary sta (variable) numeric_cols: Index[str]
summary = df[numeric_cols].describe(percentiles=[.01, .05, .25, .5, .75, .95, .99]).T
summary["skew"] = df[numeric_cols].skew(numeric_only=True)
summary
```

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max	skew
Length of Stay	107682.0	5.055980	7.440349	1.00	1.0000	1.0000	2.00	3.000	5.0000	15.000	35.0000	120.00	7.435743
Year	107682.0	2018.495682	0.499984	2018.00	2018.0000	2018.0000	2018.00	2018.000	2019.0000	2019.000	2019.0000	2019.00	0.017274
Total Charges	107682.0	52578.678601	88236.674008	1500.00	4455.2673	6052.0000	17181.89	29873.045	55216.6475	168937.794	381260.6335	4409949.53	10.163514
Total Costs	107682.0	15946.164155	22726.076002	88.27	791.1400	947.5245	5410.03	9974.455	18189.6025	49996.698	103110.3964	956351.23	8.046440

Figure 1: Summary statistics for LoS, charges and costs

2.6.1.2. Distributions of Numerical Values

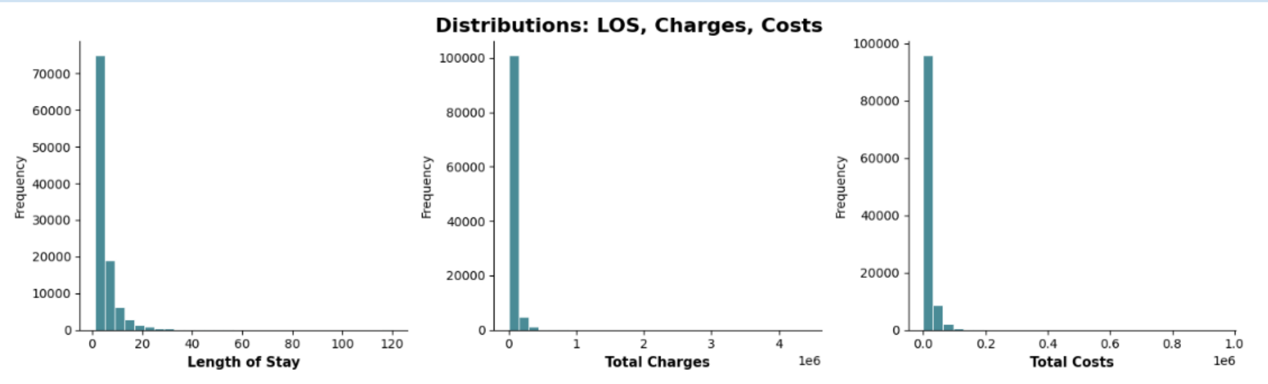

Figure 2: Distributions for LOS, Charges and Costs

LOS highly right-skewed, with a median of 3 days and a mean of 5 days. While most discharges occur within 5 days, a small number of cases extend to significantly longer durations, reflecting complex or severe conditions. Only 5% remain more than 15 day

Total Charges exhibit a highly right-skewed distribution, with a median of approximately USD 30K, and a mean of USD 54K. A small proportion of cases accounts for extremely high charges, as indicated by the 99th percentile exceeding $379,000 and a maximum above USD 4.4 million

Total Costs also demonstrate significant right-skewness, with a median of approximately USD 10K and a mean of USD 16K. Similar to total charges, a small number of cases incur substantially higher costs.

2.6.1.3. Outliers

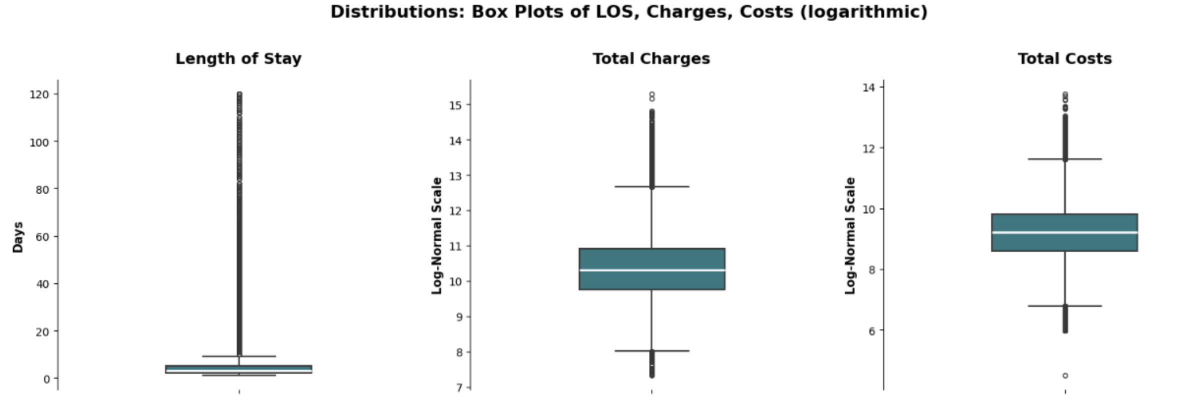

Figure 3: Outlier distributions in Box Plots for length of stay (LOS), total charges, and total costs, with total charges and total costs displayed on a logarithmic scale

The log-transformed distributions of total charges and total costs show approximately normal shapes, consistent with log-normal distributions. This pattern indicates that the extreme values observed in the raw data are not statistical anomalies, but rather intrinsic features of the healthcare business model. In particular, a small proportion of cases accounts for a disproportionately large share of overall costs. This phenomenon is documented in the healthcare literature Amorim et al. and is also clearly evident in our dataset.

Below is a figure showing the actual log-normal distribution.

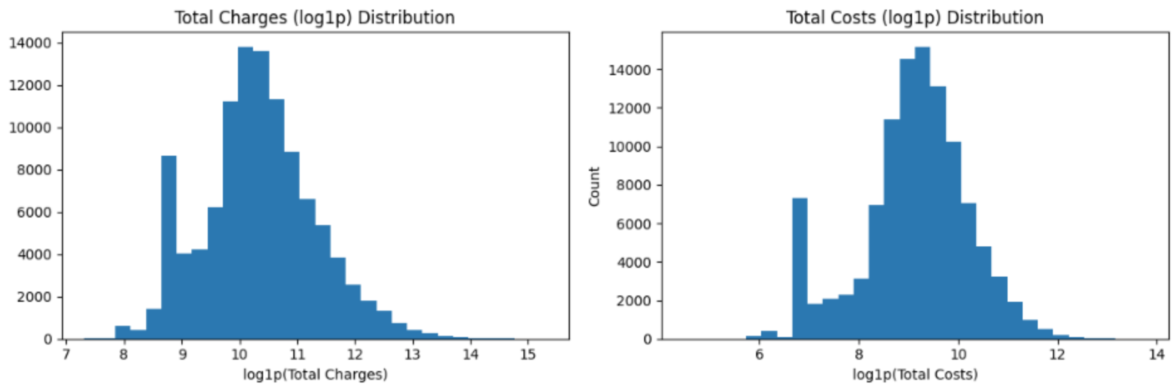

Figure 4: Logarithmic distributions of Costs and Charges for the Healthcare dataset

2.6.2. Categorical Features

Categorical Features were analysed in Count plots, so visualize how they contribute in the dataset

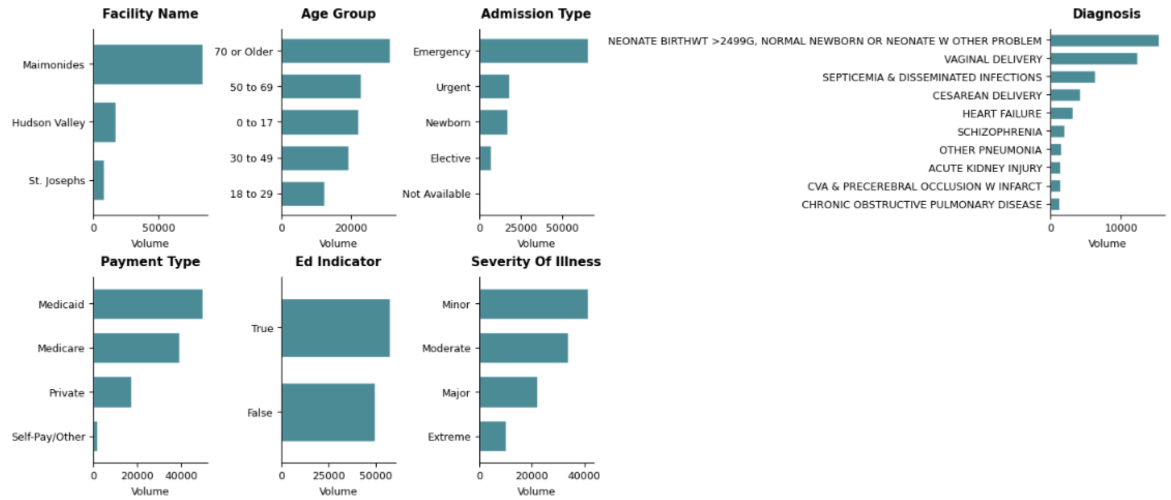

Figure 5: Distribution of hospital discharges by facility, age group, admission type, diagnosis, payment type, and emergency department (ED) indicator. Volumes represent the number of recorded inpatient cases in the sample.

The insights that we have from the above analysis, are the following:

- Maimonides dominates in cases as the largest facility.
- The patient population is predominantly aged 70 years and older, indicating that the hospitals serve a relatively older population.
- Emergency Admission dominates massively. This implies that the hospitals are forced to operate reactively
- 80% of emergency admissions end up in the Emergency Department.. (This has been found through a crosstab calculation)
- Most payments are made through Medicaid. Federal insurance for people with limited economic viability and resources.
- Most admissions are for minor severity case

2.7. Bivariate Analysis

2.7.1. Correlation Matrix

In order to incorporate ordinal categorical variables such as Age Group and Severity of Illness, we converted these categories into sequential numeric codes. This allowed us to examine their correlations with the other features in the dataset.

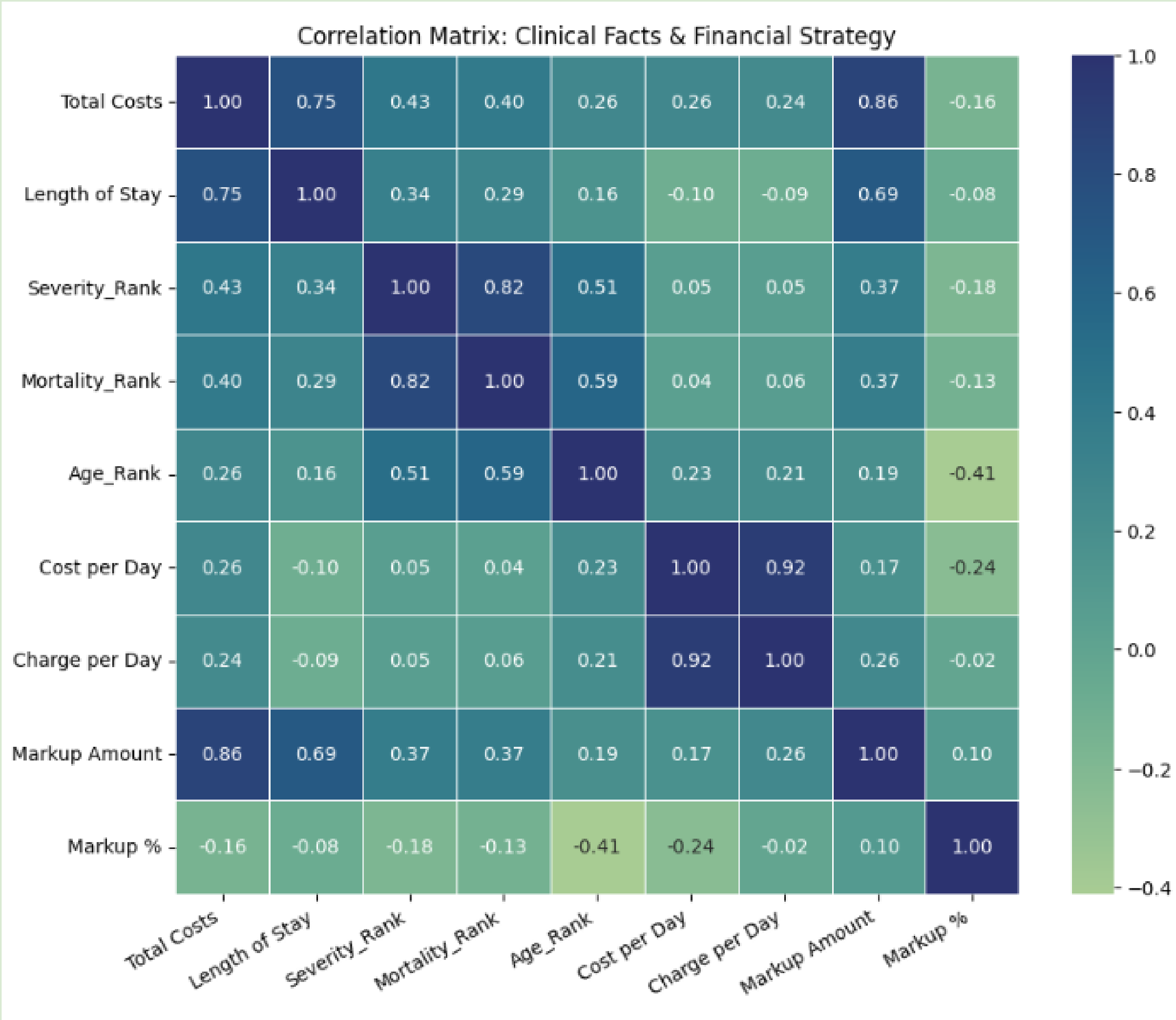

Figure 6: Correlation analysis of Numerical features and how they shape hospital revenue generation

Important insights that are derived from correlation map, drive this analysis. Specifically, this map shows:

Length of Stay (LOS) and Total Costs (corr = 0.75):

There is a strong positive correlation between length of stay and total costs, indicating that time spent in the hospital is a major driver of overall expenditures. Consequently, policies or interventions aimed at reducing LOS are likely to have a big impact on hospital financial performance.

LOS and Cost per Day (corr = −0.10):

The weak negative correlation suggests that higher daily costs tend to occur earlier during hospitalization. This pattern is consistent with more intensive and resource-needing care being delivered in the initial days of admission.

Total Costs and Markup Amount (corr = 0.86):

Total costs are very strongly correlated with the absolute markup amount, indicating that higher-cost cases generate higher absolute margins, even if percentage margins get smaller

Total Costs and Markup Percentage (corr = −0.16):

The negative correlation implies that although high-cost cases generate larger absolute markups, they tend to have lower profit margins. This suggests that efficiency gains or pricing flexibility diminish as case complexity and costs increase.

Age Rank and Markup Amount (corr = 0.19):

Patient age shows a small positive association with the absolute markup amount, indicating that older patients are associated with slightly higher revenue contributions in absolute terms.

Age Rank and Markup Percentage (corr = −0.41):

A negative correlation is observed between patient age and markup percentage, suggesting that as patient age increases, hospitals operate with a smaller profit buffer, maybe due to higher clinical complexity and constrained reimbursement.

Severity of Illness and Mortality Risk (corr = 0.82):

A strong positive correlation exists between severity of illness and mortality risk, confirming that more severe cases are associated with substantially higher mortality, as expected in clinical settings.

Overall, the correlation analysis highlights length of stay, patient age, and illness severity as key dimensions shaping hospital cost structures, revenue generation, and financial risk exposure, and these insights are used throughout this analysis

2.7.2. Scatter Plots - Regression and Clustering

The analysis of the scatterplots in the figure below, indicates that **Maimonides has a different billing pattern** compared to St. Joseph's and Hudson Valley.

At Maimonides, pricing lines are separated for the Payment Types. Particularly for Medicaid, this appears along three distinct trajectories. This suggests the presence of **payer-specific pricing structures**, likely reflecting different contractual or disease specific detailed pricing requirements.

Maimonides serves a very large Medicaid population (see figure below) and relies on **differentiated markups for insured patients**, which can create visible separation in prices for identical services. The steep slopes observed in Maimonides' pricing relationships indicate that increases in listed charges translate into substantially higher collected amounts for specific payer types.

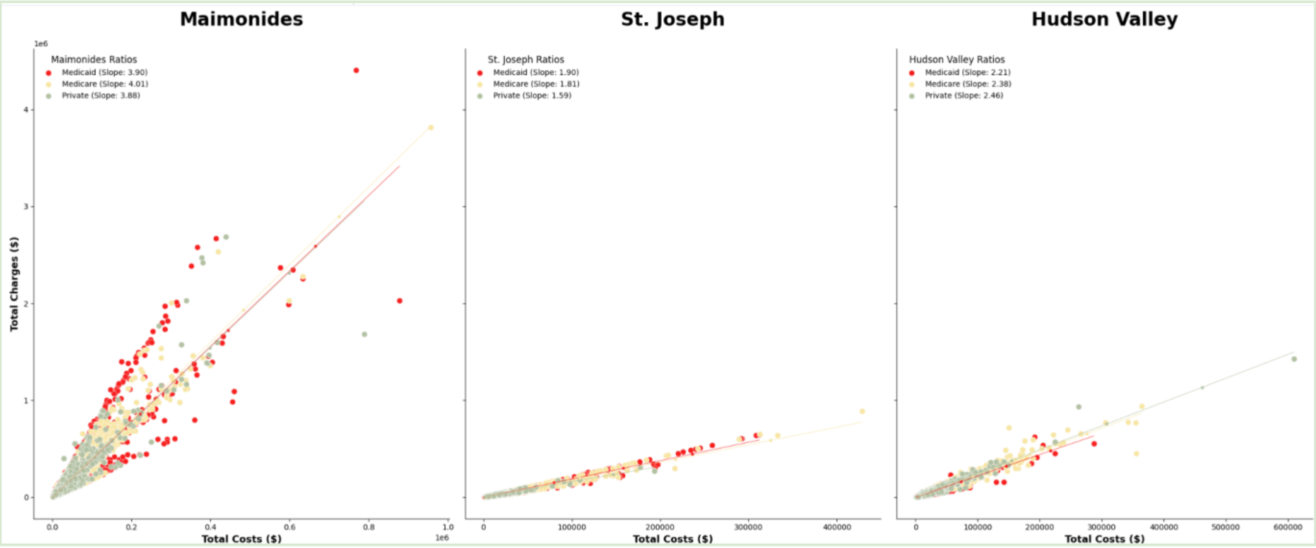

Figure 7: Scatterplots of Total Costs vs Total Charges for the 3 hospitals. Finding regression lines (multipliers) for Cost vs. Charges with linear regression.

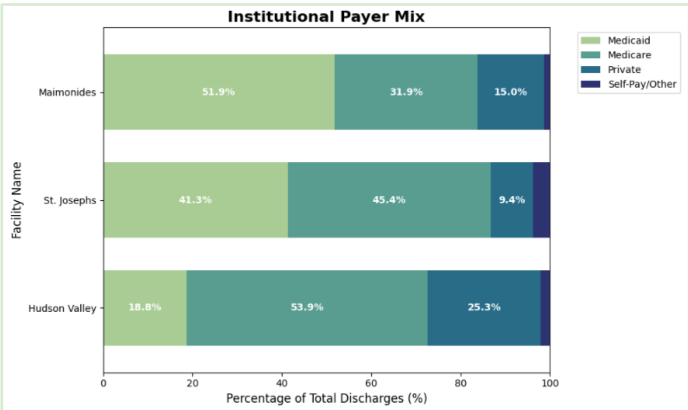

Figure 8: Payer Type mix per Hospital

In contrast, **St. Joseph's and Hudson Valley display much narrower and closely clustered pricing lines** across payer categories. This suggests the presence of a **more standardized charges**, where prices vary less by payer type. Their lower slopes imply that hospital costs result in comparatively smaller increases in actual payments.

Overall, the findings point to **two distinct billing approaches**.

- **St. Joseph's and Hudson Valley** follow a relatively *standardized* pricing model, with uniform markups across payers.

- **Maimonides** follows a more *differentiated* strategy, consistent with broader cross-subsidization behaviour and the financial pressures associated with serving a high-Medicaid population.

This tells us that hospital billing is not merely a product of insurance type, but a specific 'Institutional' issue. This proves that while the 'PayerType' determines the general revenue tier, the 'Facility' determines the absolute markup buffer, indicating a standardized but facility-specific pricing strategy.

We decided to isolate Maimonides and specifically for Medicaid Costs and Charges and run KMeans clustering for 3 distinct clusters (k = 3) based on pricing. What we found may be seen in the cluster map below.

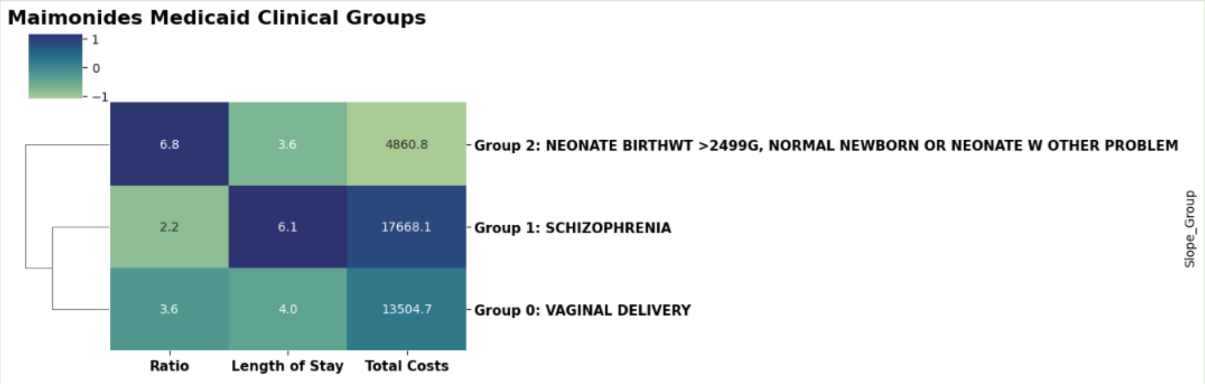

Figure 9: Cluster map of Medicaid inpatient cases at Maimonides Hospital based on K-Means clustering (k = 3), using cost-to-charge ratio

We have been able to find out what each of the clusters represent as the most frequently appearing disease, although there is diffusion of the same diseases in all 3 clusters.

Group 0 – represents "Routine Cases" or standard inpatient cases. These patients show **average length of stay**, **moderate costs**, and **mid-range CCR values**, indicating neither exceptionally high resource utilization nor unusually low pricing. Clinically, this group contains common and relatively predictable admissions (e.g., maternity-related cases), which aligns with the stable cost. From a financial perspective, this cluster reflects the hospital's baseline operational activity.

Group 1 – represents the "Loss Makers". These cases are characterized by a **very long LOS**, **high total costs**, and **low CCR values**, meaning that even large costs do not translate into increased revenue. Patients require extended, long time and complex care while reimbursement remains low under Medicaid. These admissions place substantial strain on the hospital's margins

Group 2 – represents the “Cash Cows”. These patients show **short LOS**, **low costs**, but **high CCR values**, indicating that billed charges convert to revenue more efficiently. This group creates a disproportionate share of positive margin. In operational terms, these cases can be considered **high-efficiency or high-yield**, where limited resource use still results in relatively large financial outcomes and profit margins.

This segmentation supports the broader finding that **facility-specific institutional practices**—rather than payer type alone—shape financial performance. The clusters highlight how Maimonides is balancing high Medicaid demand with selective lines of efficient charging to remain financially viable.

2.8. Forecasting Charges for the Next Decade

The dataset covers only a two-year time span, which introduces limitations for long-term forecasting using linear regression. The projections of charges and costs over a ten-year horizon should be interpreted with caution. These forecasts are more of an exercise rather than depicting a true analysis as they are unlikely to capture the reality and market dynamics.

The results show the variation in projected financial performance across hospitals.

St. Joseph’s is forecasted to experience the lowest annual margin expansion, which coincides with the high average length of stay. In contrast, Maimonides exhibits the largest projected margin expansion per year, with lower average LOS, indicating a more favourable conversion of estimated costs to revenue. Hudson Valley falls between the two, showing moderate cost growth and margin expansion.

We may find from this analysis the revenue expansion and how much it differs per institution. These differences are summarized in the table below.

Table 3: Showing the Profit Margin Expansion for each institution

Hospital	Cost Increase/yr	Revenue Increase /yr	Margin Expansion /yr	Avg LOS
St. Josephs	$2,457	$3,113	$656	7.77
Maimonides	$1,463	$3,967	$2,503	5.00
Hudson Valley	$469	$1,796	$1,327	4.05

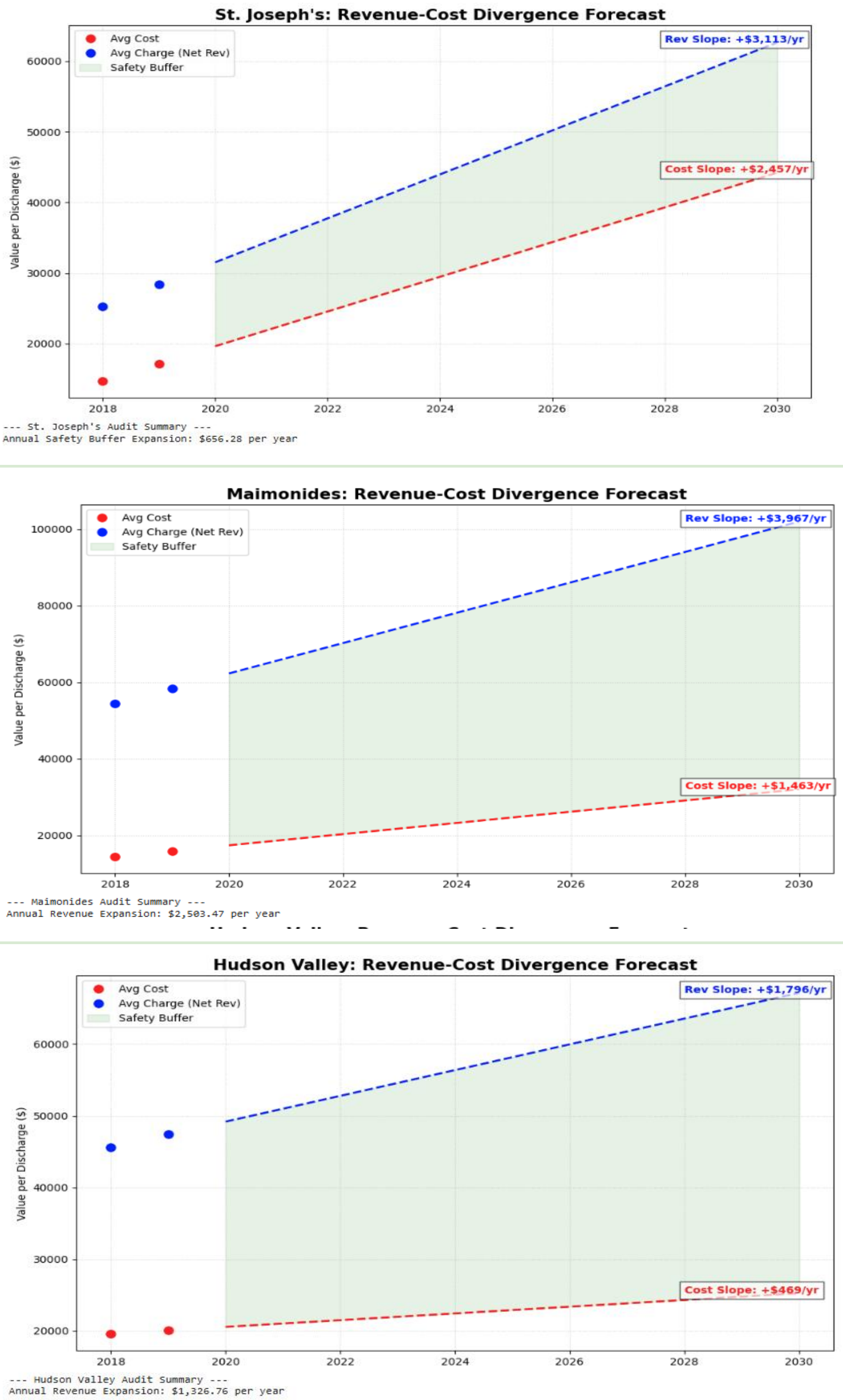

Figure 10: projected trajectories of hospital charges over a ten-year period. We highlight the differences in revenue expansion patterns across institutions.

3. Dashboards in Power BI Desktop

Power BI Desktop is used for creating interactive dashboards since it is more reliable than the BI Service counterpart. BI as a Service is more limited and it is mainly used for sharing and publishing dashboards that have already been created locally using BI Desktop.

Some of the reasons BI desktop is preferred, in comparison with BI as a service are (Datacamp):

1) Complete access to Power Query Editor.
2) Complete access to Data Modelling features
3) Local control is, in most cases, a better option than cloud control. Most professional workflows are commenced locally and published later on.

3.1. Import Data in Power Bl Desktop

To import data into Power Bl Desktop (from now on: Dashboarding Tool, or simply the

tool), follow these steps:

1) Launch the tool and create a new blank report.
2) Click Home → Get Data Text/CSV.
3) Select the dataset to upload.
4) Power Query will launch automatically. Make sure the Transform option is selected.

3.2. Data Import Integrity

3.2.1. Number of Rows and Columns

The number of rows (53,376) and columns (18) has been validated in Power BI.

The row count can be seen by selecting Table view and looking at the bottom, where the dataset name is displayed along with the row number in parentheses.

To see the columns, select Home → Transform Data. Power Query will open, and the number of columns is displayed at the bottom of the window.

3.2.2. Check for Nulls, Corrupted Values and Data Types

To check for null values:

1) Open Power Query

2) In the View tab, enable Column distribution and Column profile to display additional information at the bottom of the table. This includes details such as errors in the dataset and empty cells, among others.
3) At the same panel, at the column headers, the data types can be seen as well.

3.2.3. Check for Duplicates

1) Open Power Query and select the Home tab.
2) Then select Remove Rows → Remove Duplicates.

If the number of rows remains unchanged after this action, it indicates that no duplicate records exist. In this dataset, the number of rows is reduced, indicating the presence of 2,631 duplicate rows. However, these duplicates correspond to valid observations (e.g., repeated categories or similar records) and should not be removed, as this would lead to loss of important information.

3.3. The Technical Dashboard

The technical dashboard is included in the Technical.pbix file, and also shown here as a screenshot.

It serves as the deep data analysis of the study, showing results into granular, data-driven evidence. It uses key operational, financial, and utilization metrics to support institutional comparisons and to validate earlier findings in the EDA regarding billing strategies, profit efficiency, and cost structure.

3.3.1. KPIs (Top Panel)

The dashboard starts with a horizontal list of KPIs that show an operational baseline. The average Charge-to-Cost Ratio (CCR) of 3.65 suggests that, on average, charges are approximately 3.65 times higher than costs. While this shows as a good indication, 30.1% of cases are classified as high-severity (Major or Extreme), relating to intensive resource utilization that inflates both costs and charges.

Also, the emergency admission rate of 53.7% highlights a reactive operating mode. High emergency admission rates are a struggle for planned resource allocation and translates to higher per-day costs. This is reflected in the average cost per day of $3.76K, which has increased by 15.2% since 2018, suggesting growing operational pressure.

Finally, the Cross-Subsidy Ratio (CSR) of 1.02 indicates a balance between high-margin and loss-making cases. This implies that, at a high level, where we aggregate for all institutions for all years, the surplus from efficient cases is sufficient to cover for losses from high-complexity admissions.

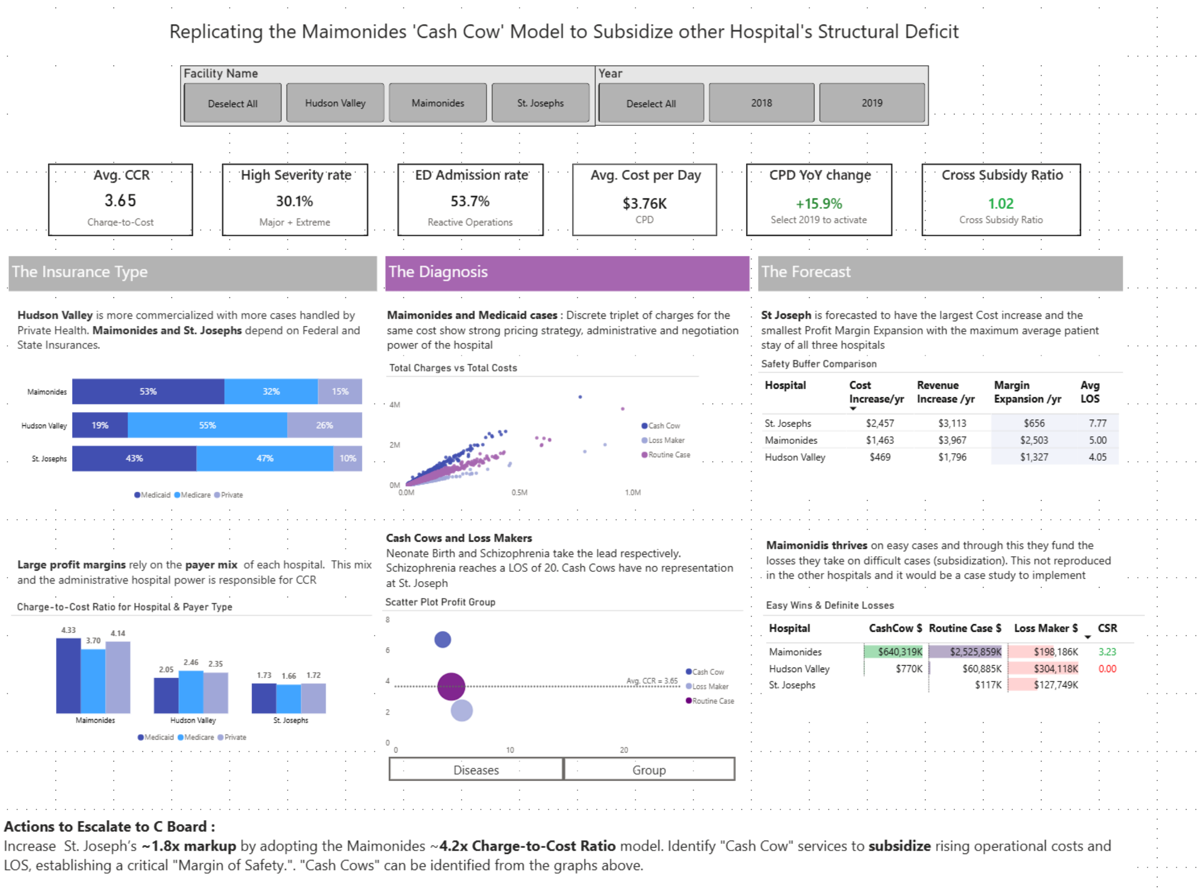

Figure 11: A screenshot of the technical dashboard.

3.3.2. Graphs

Left Column (Insurance Type and Payer Segmentation)

The payer mixed analysis reveals institutional heterogeneity. Hudson Valley has a more commercialized profile, with approximately 26% of cases covered by private insurance. In contrast, Maimonides and St. Joseph's rely heavily on public payers, with Medicaid accounting for up to 53% of total volume.

Beyond volume differences, the dashboard also shows how hospitals vary in their use of charges by payer type. St. Joseph's applies relatively limited and low in CCR charge differentiation, suggesting a more homogeneous pricing strategy. Maimonides, however, demonstrates substantially higher CCR multipliers by payer, indicating more aggressive and differentiated utilization of billing across insurance categories and specifics of diseases.

At the diagnosis level, the dashboard reveals the “discrete triplet” pricing structure within Maimonides’ Medicaid cases. Charges cluster into three distinct tiers rather than forming a compact distribution as happens with Hudson Valley and St Joseph. This pattern suggests structured administrative categorization of diagnoses into predefined pricing bands, likely reflecting contractual, regulatory, or internal billing rules.

Further insight is obtained through K-Means clustering, which identifies the financially distinct clinical groups.

Neonate births emerge as high-efficiency cases, characterized by relatively low costs and high cost-to-charge ratios, effectively functioning as “Cash Cow” cases. On the other hand, psychiatric diagnoses such as schizophrenia are associated with extended length of stay, reaching up to 20 days at St. Joseph’s, while generating limited charges. These cases constitute persistent loss makers unless cross-subsidization works for the institution. Drill through capabilities, show which diseases fall under which category.

Right Column (Forecasting and Financial Buffer Assessment)

Linear regression estimates suggest that St. Joseph’s faces the largest annual cost increase ($2,457 per year) while experiencing the lowest projected profit margin expansion ($656 per year). This combination implies a narrowing financial buffer over time, increasing exposure to any problem that might arise

In contrast, Maimonides is forecasted to expand its margin by approximately $2,503 per year, reflecting a strong pricing efficiency. The institution’s ability to generate revenue from “Cash Cows” cases provides the ability to absorb losses from Loss Maker Cases. Maimonides seems to be very good at it with an individual Cross Subsidy Ratio of 3.23 above the average

The technical dashboard demonstrates that financial outcomes are not incidental but reflect institution-specific optimization strategies. Maimonides’ pricing structure, with high CCR on efficient cases, enables sustained cross-subsidization. These findings reinforce the core conclusion that facility-level institutional behaviour, rather than payer type alone, determines the financial performance.

The technical dashboard can also be used to identify specific invoicing SKUs associated with the observed pricing strategies, and highlighting elements that could potentially be replicated across other institutions.

3.4. The Director Dashboard

The director dashboard can be found in the Director_and_C.pbix file, in the sheet named “Director-Level.”

The audience of this dashboard is of intermediate level of hierarchy. In some cases, intermediate levels can act as top levels in a way where they do not need to communicate further information to the top level, since they have enough power to tackle many problems or apply many optimizations by themselves.

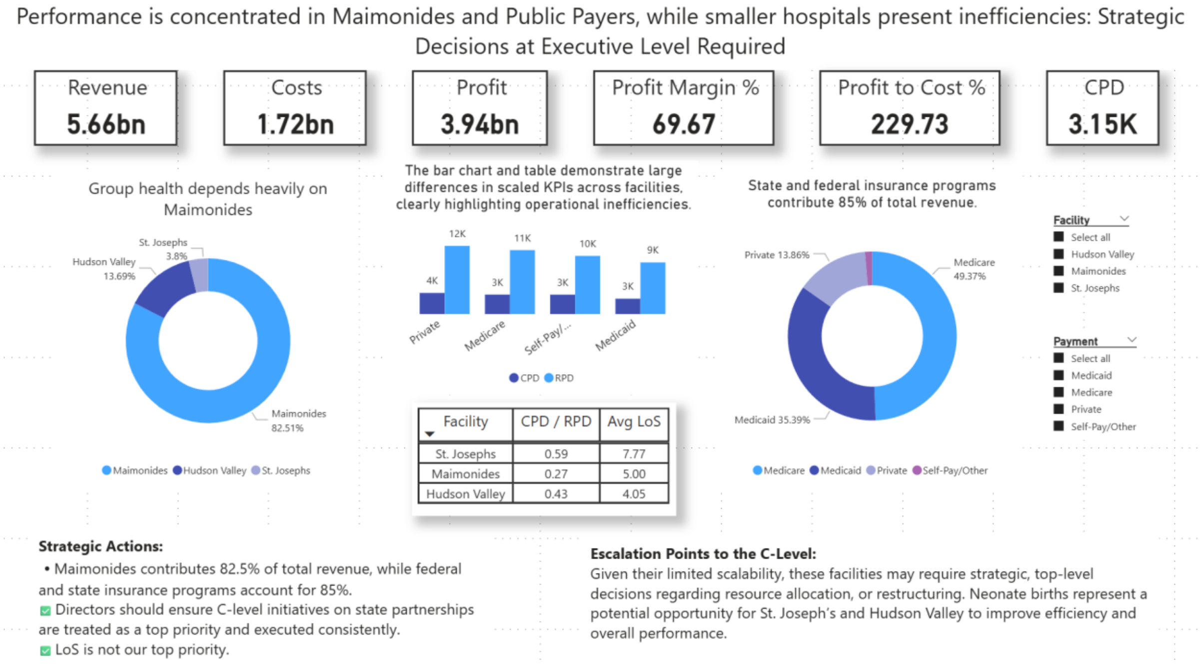

Figure 12: A screenshot of the director dashboard.

3.4.1. KPIs (Top Panel)

There is a strategic selection of KPIs for this level of hierarchy, containing all three top financial KPIs, that is, revenue, costs, and profit, while there are three normalized ratios included as well, that is, profit margin, cost-to-profit, and cost per patient per day (CPD). This final KPI is both normalized and scaled so that it is displayed in thousands, instead of billions, assuming human everyday activities are related mostly to transactions in tens to thousands of dollars, rather than billions of dollars.

Concerning the remaining five KPIs, and in combination with the facility slicer, they can express what is financially happening with the hospitals in many details, and in a quick way, which fits intermediate hierarchy levels.

It can be seen that Maimonides is the winner and the most efficient hospital, while there is a 50% cost-to-profit drop to Hudson Valley and a further 50% drop to St. Joseph's, and a 22% profit margin drop to Hudson Valley and a further 30% drop to St. Joseph's.

3.4.2. Graphs

Two donut charts are included, from which directors can quickly see what is happening in the group's revenue because of Maimonides and state or federal level payers.

The bar chart, along with the table in the middle, contains the normalized and scaled CPD and RPD (revenue per patient per day) for eye-friendly comparisons, along with the CPD-to-RPD ratio, which clearly shows that Maimonides is the best hospital of the group.

The table also contains the average length of stay (LoS) intentionally, to show that there is no clear positive or negative relation between the best hospital and this specific KPI. Specifically, Maimonides demonstrates neither the highest nor the lowest average LoS. This is enough evidence for the intermediate level not to communicate this KPI importance further to the top level, since more detailed investigation is needed to see what this KPI contributes financially to the group. This work can be delegated to lower hierarchy levels, which will communicate the results back to the directors in the future. Therefore, in real life, this is a continuous loop of information moving.

3.5. The C-level Dashboard

The C-Level dashboard can be found in the Director_and_C.pbix file, in the sheet named "C-Level."

The audience of this dashboard is at the top level of the hierarchy. They are responsible for taking the boldest actions in the group to optimize the financial situation and tackle specific issues. This does not mean that they ignore other information, such as whether the hospital revenue flows are sufficient

3.5.1. KPIs (Top Panel)

The same number of KPIs as the director's dashboard has been included here. The top level does not need more than six important KPIs to understand what is happening, whether the hospitals are winning or losing.

Specifically, only revenue and profit are kept here, as revenue already contains the costs, and including both would be redundant for such a top level. Profit margin is one of the most important financial KPIs, so it is included as well, as it is the only KPI ratio in the dashboard that directly shows what is financially happening. CPD is included here too because it also reveals important information about inefficiencies. The top panel is completed by adding the number of discharged patients and the average LoS.

From the top panel, it can be quickly seen that Maimonides is by far the winner of the group, while the other hospitals present inefficiencies. Especially St. Joseph's showcases strange results from its operational activities, since it does not score well, although it has the smallest CPD.

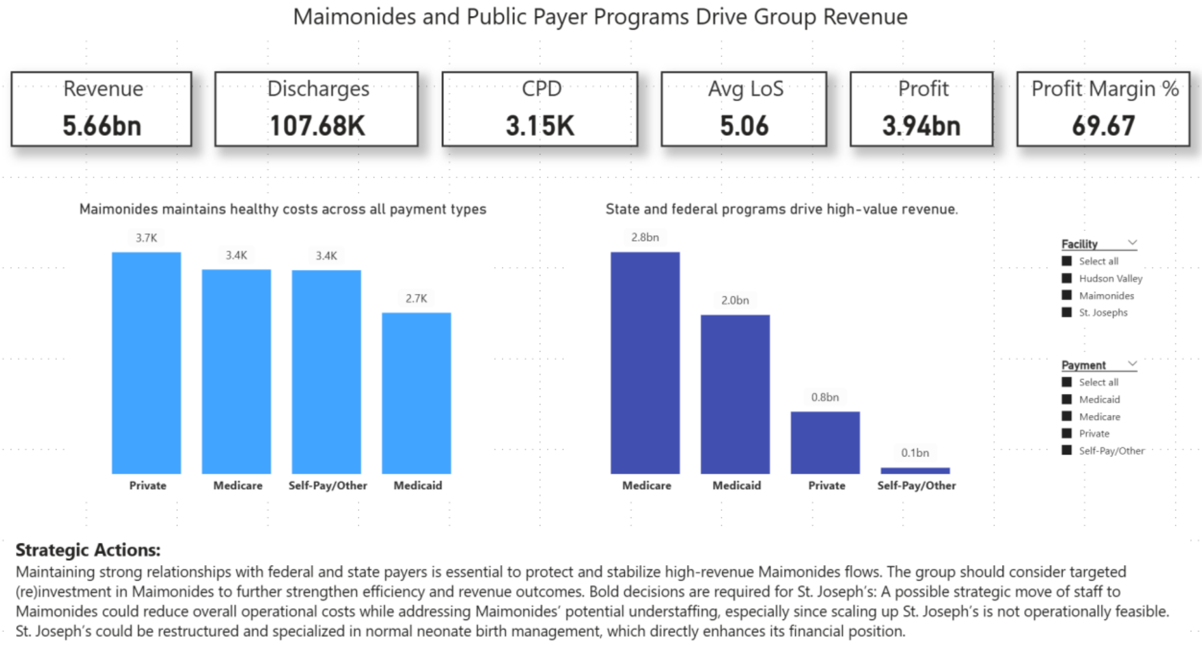

Figure 13: A screenshot of the C-level dashboard.

3.5.2. Graphs

The dashboard is completed by adding only two simple bar charts so that the top level can see the entire picture without spending more than one minute.

The bar charts show that state or federal level payers contribute tremendous amounts of revenue to the group, and therefore the group's relationships with them must be secured. Another conclusion is that Maimonides, although it does not demonstrate the lowest CPD, presents a uniform cost distribution across different payment types with deviations up to 20%. On the other hand, St. Joseph's, although it has the lowest CPD, demonstrates large standard deviations across different payment types with deviations up to 31%.

4. Conclusion & Strategic Recommendations

This project demonstrates how students' imagination can create a structured workflow, extending from data preprocessing and EDA on a real dataset to dashboard discussions across different hierarchy levels. Many tools are used, such as Python, Google Colab, GitHub, Power BI, and MS Office, while potential real-life scenarios are presented for each management level. This shows flexibility and the ability to combine tools effectively, adding real value to the Data Visualization class.

Adopting a bottom-up approach, the project concludes with some bold strategic actions related to the C-level, which holds the most power in the group:

1. (Re)Investment in Maimonides to strengthen its position within the group.
2. Securing strong relationships with the state at all costs, or even expanding them.
3. Reallocating human resources from St. Joseph's to Maimonides to address potential understaffing while reducing operational costs of St. Joseph's.
4. Restructuring St. Joseph's to host highly profitable medical cases, such as normal neonate births, to directly improve revenue

It would be extremely interesting to monitor how these recommendations perform in real life and how they would affect the group. Of course, this monitoring would require dashboards as well, showing how the importance of this class continues in a continuous loop of information across hierarchy levels.

Citations

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(3) Maimonides Site: https://maimo.org/

(4) Hudson Valley Site: https://www.nyp.org/locations/westchester-hudson-valley/nyp-hudson-valley-hospital

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